

Explaining Variation in Fertility Rates Through Socioeconomic Indicators

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1. Abstract:

The average number of children a woman is expected to have in her lifetime is a commonly used metric for assessing birth rates. As it plays a central role in determining a country's long-term demographic structure, it is often used to understand population dynamics. By tracking fertility trends, governments can anticipate future challenges and opportunities through planning and policy-making in areas like healthcare systems, resource allocation, education and social welfare programs.

Across the world, countries display distinct fertility patterns, reflecting the complex interaction of cultural norms, social expectations, and economic conditions influencing family planning policies worldwide. Although economic growth is typically associated with lower fertility levels, noticeable variations among countries at similar stages of development suggest that economic factors alone are insufficient in fully accounting for these trends. Social factors, including education levels, healthcare access, gender equality, may also significantly influence fertility behavior.

Against this background, this research project aims to investigate the extent to which socioeconomic indicators can statistically predict a nation's fertility rate, thus providing pertinent demographic insights into how they influence population structure.

2. Dataset cleaning and visualization:

For the analysis, eight socioeconomic factors potentially influencing the fertility rate of a country were selected. The chosen year for analysis was 2023, since it was the most recent year with available data on a sufficient number of countries. After locating each data set, all the gathered data was merged into a single CSV file ("merged_fertility_project_2023.csv"). Following are descriptions of the variables used in the research with their corresponding abbreviations for the RScript:

Fertility (Fertility Rate): average number of live births per woman at the end of their reproductive period if they were subject during their whole lives to the fertility rates of a given year and not subject to mortality. Data sourced from the United Nation's World Population Prospects from 2024.

Edu (Women's Education Level): average number of years women aged 25 and older have spent in formal education. Data sourced from the United Nation's Human Development Report from 2025.

GDP (Gross Domestic Product): GDP expressed in international dollars at 2021 prices and adjusted for inflation and differences in living costs between countries. Data sourced from Penn World Tables from 2025.

Femlabor (Female Labor Force Participation Rate): share of the female working-age population who are economically active. Data sourced from the International Labour Organization's Modelled Estimates database and the World Bank's World Development Indicators from 2025.

Fempolpart (Women's Political Empowerment Index): index from 0 to 1 (most empowered), measuring the extent to which women enjoy civil liberties, can participate in civil society, and are represented in politics. Data sourced from the Varieties of Democracy project from 2025.

Femlifeexp (Women's Life Expectancy): expected age of death of newborn females (female period life expectancy at birth). Data sourced from the United Nation's World Population Prospects from 2024.

Healthcare (Healthcare Expenditure): percentage of the GDP spent on public health. Data sourced from Organisation for Economic Co-operation and Development's publications from 2025.

Urbanization (Urbanization): number of people living in urban areas. Data sourced from the World Bank and the United Nations World Urbanization Prospects from 2025

Crime (Crime Rate): crime index indicating how much crime happens in a country. Data sourced from Kaggle.

The dataset was initially composed of 120 countries. Subsequently, to gain a visual understanding of the relationships between the variables, the dependent variable was plotted against each of the eight independent variables (see Appendix). This preliminary analysis allowed for the more accurate formulation of hypotheses for this study prior to performing the statistical testing and regression. The scatter plots revealed a negative correlation between healthcare expenditure, women's education level, and life expectancy. Unsurprisingly, crime rate was positively correlated to our dependent variable, whereas urbanization and GDP showed no connection to fertility rates. As for

female labor force participation rate and women's political empowerment index, they had a weaker negative correlation with the dependent variable.

Furthermore, three countries were identified as outliers through interactive plots and thus excluded from the final dataset analysis to prevent possible interference. Accordingly, after the necessary cleaning, it consisted of 117 countries ("merged_fertility_project_2023_final.csv").

3. Multivariate Linear Regression:

As aforementioned, the aim of the project is to predict the response variable (fertility rate) with the explanatory variables (healthcare expenditure, education levels, women's empowerment index, life expectancy, crime rate, urbanization, GDP, and female labor force participation rate). Thus, a multivariate linear regression was the chosen model for the purposes of this study (see Appendix for R script). GDP and urbanization are included in logarithmic form to account for non-linear effects and scale differences, improving the regression's statistical properties.

Analyzing the p-value of each test, women's education level, healthcare expenditure, and women's political empowerment index do not show a statistically significant association with the dependent variable. Though the preliminary scatterplot revealed some individual correlation, once the other variables were controlled in the regression, the effect disappeared. Meanwhile, log urbanization and female life expectancy are both below the 0.05 significance level and thus are relevant predictors of fertility rate. Notably, female life expectancy plays a substantial role for predicting fertility rates with its p-value being exceedingly below the 0.05 level. At the 0.1 level, crime rate, log GDP, and female labor participation display a marginal statistical significance as they have a weak association to the dependent variable.

Around 70.2% of the variation in fertility rates across countries is explained by the response variables, as indicated by the multiple R-squared value of 0.7022. After accounting for the number of predictors in the regression model, the adjusted R-squared value is still high at 0.6795 which suggests that the model is well specified and not overfitted. Together, these results show strong explanatory power and support the suitability of the model. Overall, the explanatory variables are jointly significant and that the model provides a statistically meaningful explanation of fertility rates given that the F-statistic is highly significant ($p < 0.001$).

4. Model Selection:

Next, nested model comparison was employed to determine whether there is a simpler model that explains the variations in the response variable with less covariates. Given that women's education level, healthcare expenditure, and women's empowerment index were deemed not statistically significant predictors of fertility rate, they were the covariates selected to be excluded from the model. The null hypothesis considers that the three variables do not add exploratory power once added into the model, accordingly: $H_0: \beta_{\text{Education}} = \beta_{\text{Healthcare}} = \beta_{\text{FemalePolitical}} = 0$.

The test revealed an F statistic with a large p-value of 0.36 that fails to reject the null hypothesis, suggesting that there is no statistical evidence of these covariates improving the model. Therefore, it indicated that their exclusion allows for a simpler model of the fertility rate.

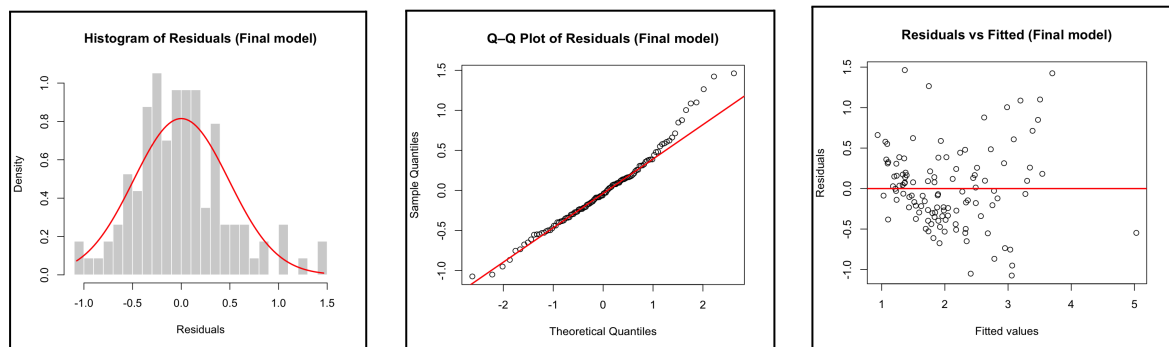
Model 1: Fertility_rate ~ log_GDP + Female_labor_participation + Female_life_expectancy + log_Urbanization + Crime_Rate						
Model 2: Fertility_rate ~ Education + log_GDP + Female_labor_participation + Female_life_expectancy + Healthcare + log_Urbanization + Female_political_participation + Crime_Rate						
	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	108	27.071				
2	105	26.267	3	0.80408	1.0714	0.3646

To complement the nested model comparison, an additional model selection method that considers different criteria was employed. Starting from the multivariate regression model that includes all eight initial covariates, backward stepwise selection using the Akaike Information Criterion (AIC) was applied (see Appendix for Rscript). This was done to allow for model selection to be based on the combination of variables that provides the best trade off between fit and complexity rather than individual coefficient significance.

The resulting model retains only five covariates (log GDP, female labor force participation, female life expectancy, urbanization, and crime rate) while maintaining a comparable adjusted

R-squared (0.6789 vs 0.6795), indicating that the excluded variables do not greatly improve model performance once other factors are accounted for. Interestingly, even though this method did not iteratively remove variables based on statistical significance, it still excluded the same covariates as those identified through the significance-based nested model comparison.

In order to verify the assumptions of the multivariate linear regression, an analysis of its output was carried out. From the Residuals Histogram, it is evident that the residuals are centered around zero and display a slightly asymmetric distribution. This supports what is visually indicated by the Normal Q-Q plot: The residuals closely follow the diagonal with only small deviations at the tails. An additional numerical analysis was done using the Shapiro-Wilk test, which suggested a statistically significant deviation from normality for the residuals. However, it is debatable whether this indication is practically significant. Also, the Residual vs Fitted plot proposes a mild heteroskedasticity, but displaying no noticeable funnel shape. Thus, it is not a severe enough violation of homoscedasticity to undermine the validity of the model.



5. Statistical Testing:

Though education appeared to have some theoretical correlation to the dependent variable, the results of this experiment have thus far indicated that it has no statistical significance for explaining the changes in the fertility rate when considering other socioeconomic factors. This surprising discovery led to the desire to further investigate the matter, conducting a statistical testing model that explored the question: “Do countries with higher female education levels have different fertility rates from those with lower ones?”

For this analysis, only the information concerning education and fertility rates from the final version of the dataset was necessary. The countries were split into high and low education groups by calculating the median, resulting in a boxplot as a preliminary visual exploration (see Appendix). Furthermore, both the Welch Two-Sample t-test and the Wilcoxon rank-sum test were employed. Both are well suited for the goals of the study, however, while the former assess differences in average fertility rates and allows for unequal variances, the latter considers systematic variations in the dependent variable and doesn't rely on normality assumptions. Together, they provide a robust framework for statistical inference.

Firstly, the Welch Two-Sample t-test was carried out using the following hypotheses:

H0: The average fertility rates are the same for high and low education countries.

H1: The average fertility rates differ for high and low education countries.

Welch Two Sample t-test		
data: Fertility_rate by Edu_group		
t = -5.6824, df = 84.271, p-value = 1.856e-07		
alternative hypothesis: true difference in means between group High education and group Low education is not equal to 0		
95 percent confidence interval:		
-1.1053062 -0.5322514		
sample estimates:		
mean in group High education	mean in group Low education	
1.632169	2.450948	

As seen above, it resulted in a very small p-value, thus rejecting the null hypothesis and affirming the initial belief that there is statistically significant evidence that countries with high education levels exhibit a lower average fertility (mean=1.63) rate than countries with lower education levels (mean=2.45).

Secondly, the Wilcoxon rank-sum test was carried out using the following hypotheses:
H0: fertility rates are systematically the same for high and low education countries
H1: fertility rates systematically differ for high and low education countries

```
Wilcoxon rank sum test with continuity correction  
  
data: Fertility_rate by Edu_group  
W = 737, p-value = 1.114e-07  
alternative hypothesis: true location shift is not equal to 0
```

This second test also resulted in a very small p-value, rejecting the null hypothesis and suggesting that there is significant statistical evidence that fertility rates are systematically lower in high-education countries. Therefore, the conclusion that the education levels are a statistically significant indicator of the fertility rate when measured independently is robust as both tests revealed few outliers and hardly any skewed data.

6. Discussion:

This project aimed to explore the extent to which socioeconomic indicators can statistically predict a nation's fertility rate. The results of this investigation showed statistical evidence of the fact that a suitable model for the dependent variable can be made from the covariates, although a limited one. However, it must be highlighted that the results are associational, not causal.

The samples relied on observational data for several countries during one specific year, leading to a study that doesn't explore temporal dynamics. Also, they were limited by data availability, and consequently not all countries were included. This implies results with a possible selection bias and a limited generalizability beyond the sampled countries. Also, relevant variables (e.g., culture, religion, access to contraception) were not included in the analysis, creating omitted variable bias and leading to a model that may not fully represent the real-world mechanisms driving fertility. Moreover, using the national average hides country-wide variations and individual-level effects, leading to aggregation bias and further limiting the generalizability of the model.

The initial multivariate linear regression including all socioeconomic covariates showed that fertility rates are strongly associated with overall levels of socioeconomic development, with effects generally aligning with theoretical expectations from the initial visual exploration. However, several covariates exhibited weak or unstable associations once included simultaneously, reflecting the high correlation among socioeconomic indicators and the difficulty of isolating their independent effects. In addition, the regression relies on key assumptions of linearity, independence, homoscedasticity, and approximately normal errors; violations of these assumptions may affect the reliability of coefficient estimates and statistical inference. Given that countries could be influenced by regional, cultural, and/or historical similarities, they may breach the assumption of independent observations.

Another potential limitation of this analysis is that the regression residuals are not perfectly normally distributed, showing a slight skewness to the right. Although the Shapiro-Wilk test confirms the lack of normality, the diagnostic plots show only a mild deviation. Therefore, given the large sample size, histogram of residuals and QQ-plots, this deviation from normality, although existent, is practically not significant and is unlikely to have a meaningful impact on the validity of the results.

To evaluate model complexity, nested model comparison and backwards AIC were used. Notably, the variables retained by the AIC procedure corresponded with those identified through significance-based nested model comparison, suggesting strong results that are not sensitive to the choice of selection method. However, while model selection yields a simpler and more concise model, it does not eliminate uncertainty regarding model choice.

The lack of statistical significance for education in the multivariate model likely reflects its correlation with other socioeconomic covariates, limited statistical power, and the possibility that education influences fertility indirectly rather than through a direct independent effect. Still, this unexpected finding prompted additional statistical testing to examine the independent relationship between education and fertility without adjustment for other covariates. Two types of two-sample hypothesis tests were used to guarantee results that remain consistent under different modeling frameworks. The Welch two-sample t-test was appropriate for this investigation as it addresses mean differences under relaxed variance assumptions, and so was the Wilcoxon rank-sum test as it compares the central tendency of two groups based on ranked observations rather than raw values.

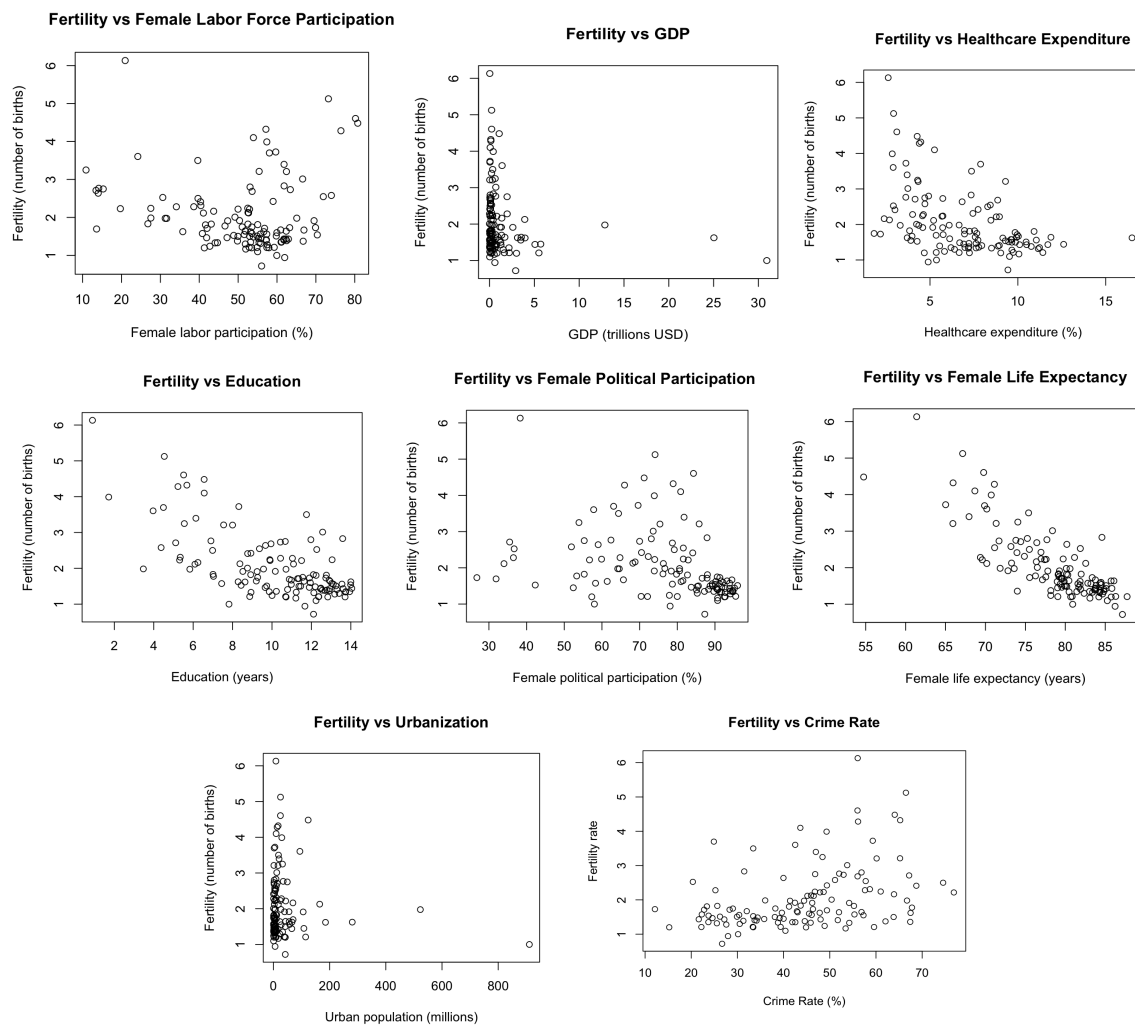
Overall, both tests suggested significantly lower fertility rates in higher-education countries, and their agreement strengthens confidence in this result.

Further research could extend this study by including longitudinal data to assess temporal variations and additional explanatory variables to potentially offer deeper insight into the mechanisms driving the fertility rate. A sub-national investigation could also be of interest as it explores within-country differences that are hidden by aggregated national measures. Additionally, future work could examine how major events (e.g., wars, pandemics, and economic recessions) affect fertility trends over time.

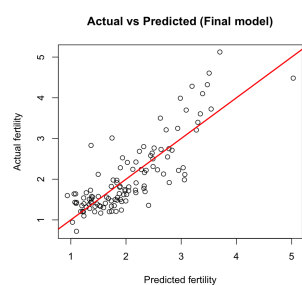
7. Appendix:

A: (Additional Graphs)

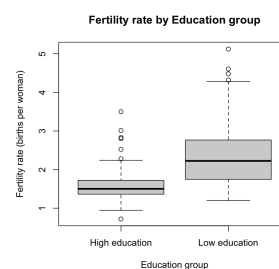
(Initial Plotting - Full Model)



(Multivariate Linear Regression)



(Initial plotting - Statistical Testing)



B: (Additional R-output)

```
Call:
lm(formula = Fertility_rate ~ Education + log_GDP + Female_labor_participation +
  Female_life_expectancy + Healthcare + log_Urbanization +
  Female_political_participation + Crime_Rate, data = final_clean)

Residuals:
    Min       1Q   Median       3Q      Max
-1.16888 -0.33553 -0.04743  0.26836  1.44067

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    12.279878   1.293748   9.492 8.48e-16 ***
Education       -0.028120   0.026684  -1.054  0.2944
log_GDP         -0.228237   0.118830  -1.921  0.0575 .
Female_labor_participation -0.006828   0.003730  -1.831  0.0700 .
Female_life_expectancy -0.090259   0.014829  -6.086 1.91e-08 ***
Healthcare      -0.033367   0.026469  -1.261  0.2103
log_Urbanization  0.238558   0.118961   2.005  0.0475 *
Female_political_participation 0.356956   0.383277   0.931  0.3538
Crime_Rate      -0.007685   0.004496  -1.709  0.0903 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5002 on 105 degrees of freedom
(3 observations deleted due to missingness)
Multiple R-squared:  0.7022,    Adjusted R-squared:  0.6795
F-statistic: 30.95 on 8 and 105 DF,  p-value: < 2.2e-16
```

(Linear Multivariate Regression with all factors)

```
Call:
lm(formula = Fertility_rate ~ log_GDP + Female_labor_participation +
  Female_life_expectancy + log_Urbanization + Crime_Rate, data = df_cc)

Residuals:
    Min       1Q   Median       3Q      Max
-1.0741 -0.3263 -0.0504  0.2532  1.4634

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    13.003620   1.219279  10.665 < 2e-16 ***
log_GDP         -0.227868   0.109937  -2.073  0.0406 *
Female_labor_participation -0.007830   0.003409  -2.297  0.0235 *
Female_life_expectancy -0.102770   0.012518  -8.209 5.17e-13 ***
log_Urbanization  0.240658   0.109613   2.196  0.0303 *
Crime_Rate      -0.007145   0.004377  -1.632  0.1055
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5007 on 108 degrees of freedom
Multiple R-squared:  0.6931,    Adjusted R-squared:  0.6789
F-statistic: 48.77 on 5 and 108 DF,  p-value: < 2.2e-16
```

(Backward Stepwise Regression)

C: (Bibliography)

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